

Adaptive Local Differential Privacy for Heterogeneous Federated IoHT Intrusion Detection

Project Proposal

Jess Alencaster, McCay Ruddick, Helen Huy

CSC429 Professor Fang

Background

Internet of Things (IoT) devices are increasingly deployed across industrial, consumer, and especially healthcare settings. Internet of Healthcare Things (IoHT) devices, such as wearable monitors and connected diagnostic systems, continuously generate and transmit sensitive network traffic data. Because IoHT devices operate in safety-critical environments, they present attractive targets for a wide range of cyberattacks. To detect such attacks, Intrusion Detection Systems (IDS) are used to monitor IoT and IoHT traffic and identify malicious behavior before it can disrupt device operation or compromise patient data. In this setting, machine learning (ML) and deep learning (DL)-based IDS have shown promise because they can learn complex patterns from large volumes of network traffic data and distinguish benign activity from attacks more effectively than many rule-based approaches.

Traditional centralized IDS approaches send raw data to a central server, which raises privacy concerns for sensitive IoHT environments. Federated Learning (FL) addresses this issue by allowing devices or gateways to train local models on their own data and share only model updates with a central aggregation server, combining them into a global IDS model without directly exposing raw traffic data.

However, FL systems remain vulnerable to membership inference attacks (MIAs), particularly because model updates can still encode private information about the local data from which they were computed [2]. Under white-box conditions, an adversary such as an honest-but-curious aggregator may exploit local updates to infer whether specific data points were used during training. This is particularly concerning in healthcare environments, where network traces may reveal sensitive patient-associated device behavior.

A common defense is to apply differential privacy (DP) by perturbing local model updates before they are uploaded. However, many privacy-preserving FL approaches rely on a fixed, uniform level of noise across all clients. In realistic federated IoHT settings, this assumption is often too simplistic because clients are inherently heterogeneous. Clients may differ in class balance, resource constraints, and data volume. Applying the same noise level across all devices may over-perturb some clients while under-protecting others, leading to an undesirable privacy–utility tradeoff.

In this project, we investigate adaptive local differential privacy for heterogeneous federated IoHT intrusion detection. We study whether client-specific noise adaptation can better balance privacy protection and IDS utility in the presence of heterogeneous local data. More specifically, we examine whether adaptive privacy mechanisms can reduce membership leakage while simultaneously preserving detection performance in federated healthcare intrusion detection.

Current Work

Current work in privacy-preserving FL-based IDS has largely explored differential privacy (DP) as a uniform noise mechanism. SECIOHT-FL [1] is a privacy-preserving FL framework for IoHT devices employing deep neural networks (DNNs) and convolutional neural networks (CNNs), applying DP through Opacus with fixed clipping norms and fixed noise multipliers during local training. Similarly, an FL-based IDS using a 1D CNN protected with DP alongside Diffie-Hellman key exchange and homomorphic encryption also applies uniform perturbation to local updates [3].

While these approaches improve privacy protection, they treat all clients as if they require the same level of perturbation. This assumption is mismatched with FL deployments in IoHT settings, where clients are inherently heterogeneous in their local data distributions, sample sizes, and resource constraints. Under such heterogeneity, applying the same fixed noise to every client can impose unnecessary utility loss on some clients while inadequately adapting to others.

Recent work has begun to address this limitation through adaptive local differential privacy in federated learning. AdapLDP-FL [4] proposes client-specific adaptive noise scaling and a direction-correction mechanism to improve the privacy–utility tradeoff under a fixed privacy budget. However, its evaluation is limited to image-classification benchmarks such as MNIST, CIFAR-10, and Fashion-MNIST. It remains unclear whether adaptive client-specific noise is effective in federated IDS settings.

Motivation

Fixed, uniform differential privacy assumes that all federated clients require the same level of perturbation, even though real IoT and IoHT deployments are inherently heterogeneous in their local data distributions, sample sizes, and device conditions. In federated healthcare intrusion detection, this mismatch is especially important because clients may observe very different proportions of normal and malicious traffic, and the data are often highly imbalanced. As a result, applying the same noise to every client can introduce unnecessary utility loss for some clients while failing to adapt effectively to others. This limitation motivates us to investigate whether a client-specific adaptive privacy mechanism can achieve a better privacy–utility tradeoff than uniform DP in FL-based healthcare and IoT intrusion detection. Building on prior adaptive local differential privacy methods such as AdapLDP-FL [4], we study whether adaptive client-level noise allocation remains effective in this setting, where performance must be evaluated not only by accuracy, but also by security-sensitive metrics such as recall and F1-score.

Contribution

This project proposes an adaptive client-specific differential privacy mechanism for federated healthcare and IoT intrusion detection, extending adaptive local differential privacy ideas from AdapLDP-FL [4] to heterogeneous federated IDS settings. Rather than allocating privacy protection across CNN layers, our project focuses on heterogeneous federated clients and investigates whether adaptive client-level noise can provide a better privacy–utility tradeoff than fixed, uniform DP. The contributions of this project are as follows:

- We design and implement a client-specific adaptive DP mechanism for FL-based healthcare and IoT intrusion detection. Building on AdapLDP-FL [4], our method adjusts the amount of noise added to each client update according to client-level characteristics rather than applying the same fixed perturbation to all clients.
- We construct a heterogeneous federated learning setting for intrusion detection by partitioning healthcare and IoT security datasets into unequal, non-IID client populations. This allows us to evaluate adaptive local DP in a setting that more closely reflects real federated deployments than uniform client partitions.
- We evaluate the privacy–utility tradeoff of adaptive client-level DP against fixed uniform DP baselines on the WUSTL-EHMS-2020 and ECU-IoHT datasets, measuring intrusion-detection performance using accuracy, precision, recall, F1-score, and privacy budget ϵ across varying noise levels.
- We assess whether adaptive client-specific noise improves empirical privacy protection against membership inference attacks (MIAs) while preserving intrusion-detection utility, providing a concrete comparison between uniform and adaptive DP in federated healthcare and IoT IDS.

Timeline

Date Range	Phase	Key Objectives & Tasks
Apr 27 – May 3 (<i>McCay's B-Day</i>)	Project Refocus, Literature Alignment, & Baseline Planning	• Finalize the refocused project proposal around adaptive client-specific DP for heterogeneous FL-based healthcare and IoT intrusion detection. • Review AdapLDP-FL and related FL-based IDS literature to determine how adaptive client-specific noise can be transferred into a healthcare/IoT intrusion-detection setting. • Finalize dataset selection and review preprocessing pipelines for the WUSTL-EHMS-2020 and ECU-IoHT datasets. • Define the initial federated learning pipeline and experimental environment for heterogeneous client simulation.
May 4 – May 10	Baseline Reproduction & Heterogeneous Client Construction	• Implement or reproduce a fixed uniform-DP federated IDS baseline to establish a comparison point. • Construct simulated heterogeneous federated clients by partitioning datasets into unequal, non-IID client populations. • Design client splits that vary in data volume, class balance, and attack distribution to better reflect realistic FL heterogeneity. • Document baseline performance under these client partitions using accuracy, precision, recall, F1-score, and privacy budget (ϵ).
May 11 – May 17	Adaptive DP Design & Experimental Formulation	• Formulate the adaptive client-level noise mechanism by building on the noise-scaling idea from AdapLDP-FL. • Define how each client's noise level will be adjusted in the healthcare/IoT IDS setting under heterogeneous client conditions. • Finalize the comparison plan: fixed uniform DP, AdapLDP-FL-style adaptive DP baseline, and our final adaptive method. • Design the white-box membership inference attack (MIA) evaluation protocol and identify the final privacy-utility metrics to report.

Date Range	Phase	Key Objectives & Tasks
May 18 – May 24	System Integration & Implementation	• Integrate adaptive client-specific noise scaling into the federated IDS training loop. • Implement the heterogeneous client-partitioning pipeline so each client receives different local data characteristics. • Ensure DP noise is added locally on each client before upload and that federated aggregation remains correct. • Debug the end-to-end FL system and verify stable training across both uniform and adaptive DP settings.
May 25 – May 30	Evaluation, MIA Testing, & Comparison	• Run the full FL-based IDS experiments on WUSTL-EHMS-2020 and ECU-IoHT under the selected heterogeneous client setups. • Compare fixed uniform DP, AdapLDP-FL-style adaptive DP, and our final adaptive client-level method. • Perform white-box MIA evaluation on all methods to assess empirical privacy leakage. • Analyze privacy–utility tradeoffs using intrusion-detection metrics and MIA performance.
May 31 – Jun 2	Final Analysis & Deliverables	• Generate final tables, graphs, and visual comparisons for adaptive versus uniform DP under heterogeneous FL settings. • Finalize the report and conclusions, emphasizing the role of client heterogeneity in motivating adaptive local DP. • Prepare the slide deck and presentation materials for the final report.

References

- [1] Fatemeh Mosaiyebzadeh, Seyedamin Pouriyeh, Meng Han, et al. “Privacy-Preserving Federated Learning-Based Intrusion Detection System for IoHT Devices”. In: *Electronics* 14.1 (2025), p. 67. URL: <https://www.mdpi.com/2079-9292/14/1/67>.
- [2] Milad Nasr, Reza Shokri, and Amir Houmansadr. “Comprehensive Privacy Analysis of Deep Learning: Passive and Active White-box Inference Attacks against Centralized and Federated Learning”. In: *2019 IEEE Symposium on Security and Privacy (SP)*. IEEE. 2019, pp. 739–753. URL: <https://ieeexplore.ieee.org/abstract/document/8835245>.
- [3] D. Torre et al. “Toward Enhancing Privacy Preservation of a Federated Learning CNN Intrusion Detection System in IoT: Method and Empirical Study”. In: *ACM Digital Library* (2025). URL: <https://dl.acm.org/doi/epdf/10.1145/3695998>.
- [4] Gaofeng Yue et al. “AdapLDP-FL: An Adaptive Local Differential Privacy for Federated Learning”. In: *IEEE Transactions on Mobile Computing* 24.6 (2025), pp. 5569–5583.